Project Demonstration

HDI Prediction System | Project Demonstration Phase

Demonstration Objective

Showcase the end-to-end working of the HDI Prediction System: from entering socio-economic indicators on the web form to receiving a predicted HDI score and its UNDP development tier.

How to Run the Application

Step	Command / Action
1	Install dependencies: pip install -r requirements.txt
2	Start the app locally: python app.py
3	Open a browser to http://127.0.0.1:5000/
4	Enter life expectancy, expected years of schooling, mean years of schooling, and GNI per capita
5	Submit the form to view the predicted HDI score and tier
6 (optional)	For production, run via Gunicorn: gunicorn app:app

Sample Walkthrough

Scenario	Life Exp.	Expected Schooling	Mean Schooling	GNI/Capita	Expected Output
Very High development	80.0	16.5	13.0	48000	HDI approx 0.90+, Tier: Very High
High development	74.0	14.0	9.5	13000	HDI approx 0.75, Tier: High
Medium development	65.0	10.0	6.5	5000	HDI approx 0.60, Tier: Medium
Low development	52.0	6.0	3.0	1000	HDI approx 0.45, Tier: Low

What the Demonstration Covers

- Live entry of the four indicators on the web form.
- Instant display of the predicted HDI score alongside its tier label.
- A deliberately invalid submission (e.g., missing field) to show graceful error handling.
- A JSON API call (e.g., via curl or Postman) to /predict, demonstrating the programmatic interface alongside the browser-based one.

Demonstration Artifacts

- Screen recording of the live web application walkthrough (to be attached separately).
- Screenshots of the input form and the resulting prediction for each development tier.
- Sample curl/Postman request-response pair for the JSON API endpoint.

Presenter Notes

Emphasize that the model was trained on real UNDP historical data (1990–2021, 195+ countries), that the tier thresholds match UNDP's own official classification bands, and that the same trained model powers both the browser UI and the JSON API — ensuring consistent predictions across access methods.